

Отчет по лабораторной работе №6

Установка и настройка системы управления базами данных MariaDB

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Цель работы

Приобретение практических навыков по установке и конфигурированию системы управления базами данных на примере программного обеспечения MariaDB.

Задание

1. Установите необходимые для работы MariaDB пакеты (см. раздел 6.4.1).
2. Настройте в качестве кодировки символов по умолчанию utf8 в базах данных.
3. В базе данных MariaDB создайте тестовую базу addressbook, содержащую таблицу city с полями name и city, т.е., например, для некоторого сотрудника указан город, в котором он работает (см. раздел 6.4.1).
4. Создайте резервную копию базы данных addressbook и восстановите из неё данные (см. раздел 6.4.1).
5. Напишите скрипт для Vagrant, фиксирующий действия по установке и настройке базы данных MariaDB во внутреннем окружении виртуальной машины server.
Соответствующим образом следует внести изменения в Vagrantfile (см. раздел 6.4.5).

Выполнение лабораторной работы

Установка MariaDB. Установка пакетов

```
PS D:\work\mavinogradova\vagrant> vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Mon Oct 5 13:16:46 2026
vagrant@server:~$ sudo -i mavinogradova
sudo: -i command not found
vagrant@server:~$ su - mavinogradova
Password:
Last login: Wed Sep 30 22:29:30 UTC 2026 on pts/1
[mavinogradova@server ~]$ sudo -i
[sudo] password for mavinogradova:
[root@server ~]$ dnf -y install mariadb mariadb-server
Extra Packages for Enterprise Linux 10 - x86_64          6.0 kB/s | 35 kB  00:05
Extra Packages for Enterprise Linux 10 - x86_64          852 kB/s | 6.7 MB 00:07
Rocky Linux 10 - BaseOS                                   299 B/s | 4.3 kB  00:14
Rocky Linux 10 - BaseOS                                   5.7 MB/s | 34 MB  00:05
Rocky Linux 10 - AppStream                                409 B/s | 4.4 kB  00:10
Rocky Linux 10 - AppStream                                1.7 MB/s | 2.7 MB 00:01
Rocky Linux 10 - CRB                                     [====] 0 B  --- ETA
```

Рис. 1: Вход на виртуальную машину server под пользователем mavinogradova, переход в режим суперпользователя и установка пакетов mariadb и mariadb-server

Установка MariaDB. Просмотр конфигурационных файлов

```
[root@server ~]# ls -l /etc/my.cnf.d
total 40
-rw-r--r--. 1 root root  42 Jun 30 00:00 auth_gssapi.cnf
-rw-r--r--. 1 root root 295 Jun 30 00:00 client.cnf
-rw-r--r--. 1 root root 763 May 23 15:01 enable_encryption.preset
-rw-r--r--. 1 root root 1460 Jun 30 00:00 mariadb-server.cnf
-rw-r--r--. 1 root root 232 May 23 15:01 mysql-clients.cnf
-rw-r--r--. 1 root root  76 May 23 15:01 provider_bzip2.cnf
-rw-r--r--. 1 root root  72 May 23 15:01 provider_lz4.cnf
-rw-r--r--. 1 root root  72 May 23 15:01 provider_lzo.cnf
-rw-r--r--. 1 root root  78 May 23 15:01 provider_snappy.cnf
-rw-r--r--. 1 root root 120 May 23 15:01 spider.cnf
[root@server ~]# cat /etc/my.cnf
#
# This group is read both both by the client and the server
# use it for options that affect everything
#
[client-server]
#
# include all files from the config directory
#
!includedir /etc/my.cnf.d

[root@server ~]# cat /etc/my.cnf.d/*.cnf
[mariadb]
#plugin-load-add=auth_gssapi.so
#
# These two groups are read by the client library
# Use it for options that affect all clients, but not the server
#

[client]

# This group is not read by mysql client library,
# If you use the same .cnf file for MySQL and MariaDB,
# use it for MariaDB-only client options
[client-mariadb]
```

Рис. 2: Просмотр конфигурационных файлов MariaDB

Установка MariaDB. Запуск и включение службы

```
[root@server ~]# systemctl start mariadb
[root@server ~]# systemctl enable mariadb
Created symlink '/etc/systemd/system/mysql.service' + '/usr/lib/systemd/system/mariadb.service'.
Created symlink '/etc/systemd/system/mysqld.service' + '/usr/lib/systemd/system/mariadb.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/mariadb.service' + '/usr/lib/systemd/system/mariadb.service'.
[root@server ~]# systemctl status mariadb
● mariadb.service - MariaDB 10.11 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-10-05 13:45:04 UTC; 11s ago
 Invocation: c21c4fb858863427db4a9c165b44af8fe
   Docs: man:mariadb(8)
         https://mariadb.com/docs/
```

Рис. 3: Запуск и включение службы mariadb, проверка её статуса

Установка MariaDB. Проверка прослушиваемого порта

```
[root@server ~]# ss -tulpen | grep mysql
[root@server ~]# ss -tulpen | grep 3306
tcp LISTEN 0      80          0.0.0.0:3306  0.0.0.0:*    users:(("mariadb",pid=12438,fd=18))
                                uid:27 ino:42114 sk:1a cgroup:/system.slice/mar
iadb.service <->
tcp LISTEN 0      80          [::]:3306   [::]:*       users:(("mariadb",pid=12438,fd=19))
                                uid:27 ino:42115 sk:24 cgroup:/system.slice/mar
iadb.service <->
[root@server ~]#
```

Рис. 4: Проверка прослушиваемого порта

Установка MariaDB. Настройка безопасности

```
[root@server ~]# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] n
You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] n
... skipping.

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
- Dropping test database...
... Success!
- Removing privileges on test database...
... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] y
... Success!

Cleaning up...
```

```
[root@server ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 11
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> |
```

Рис. 6: Вход в базу данных с правами администратора

Установка MariaDB. Список команд клиента

```
MariaDB [(none)]> \h

General information about MariaDB can be found at
http://mariadb.org

List of all client commands:
Note that all text commands must be first on line and end with ';'
?      (\?) Synonym for 'help'.
charset (\C) Switch to another charset. Might be needed for processing binlog with multi-byte cha
clear  (\c) Clear the current input statement.
connect (\r) Reconnect to the server. Optional arguments are db and host.
delimiter (\d) Set statement delimiter.
edit   (\e) Edit command with $EDITOR.
ego    (\G) Send command to MariaDB server, display result vertically.
exit   (\q) Exit mysql. Same as quit.
go     (\g) Send command to MariaDB server.
help   (\h) Display this help.
nopager (\n) Disable pager, print to stdout.
notee   (\t) Don't write into outfile.
nowarning (\w) Don't show warnings after every statement.
pager  (\P) Set PAGER [to_pager]. Print the query results via PAGER.
print  (\p) Print current command.
prompt (\R) Change your mysql prompt.
quit   (\q) Quit mysql.
rehash (\#) Rebuild completion hash.
sandbox (\-) Disallow commands that access the file system (except \P without an argument and \e)
```

Рис. 7: Список команд клиента MariaDB

```
MariaDB [(none)]> SHOW DATABASES;  
+-----+  
| Database |  
+-----+  
| information_schema |  
| mysql |  
| performance_schema |  
| sys |  
+-----+  
4 rows in set (0.008 sec)  
  
MariaDB [(none)]> exit  
Bye  
[root@server ~]# |
```

Рис. 8: Вывод списка доступных баз данных MariaDB

Конфигурация кодировки. Статус до изменения

```
[root@server ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 12
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.11.18-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          12
Current database:
Current user:            root@localhost
SSL:                    Not in use
Current pager:          stdout
Using outfile:           ''
Using delimiter:        ;
Server:                 MariaDB
Server version:         10.11.18-MariaDB MariaDB Server
Protocol version:       10
Connection:             Localhost via UNIX socket
Server characterset:    latin1
DB characterset:        latin1
Client characterset:    utf8mb3
Conn. characterset:     utf8mb3
UNIX socket:            /var/lib/mysql/mysql.sock
Uptime:                 11 min 6 sec

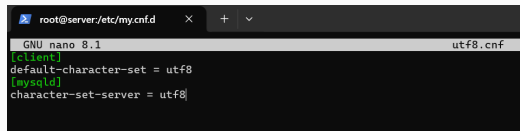
Threads: 1  Questions: 23  Slow queries: 0  Opens: 20  Open tables: 13  Queries per second avg: 0.034
-----
```

Рис. 9: Вывод статуса MariaDB до изменения кодировки

```
[root@server my.cnf.d]# cd /etc/my.cnf.d  
[root@server my.cnf.d]#  
[root@server my.cnf.d]# touch utf8.cnf  
[root@server my.cnf.d]# nano utf8.cnf  
[root@server my.cnf.d]# |
```

Рис. 10: Создание файла конфигурации utf8.cnf

Конфигурация кодировки. Содержимое utf8.cnf



```
root@server:/etc/my.cnf.d  ×  +  v
GNU nano 8.1 utf8.cnf
[client]
default-character-set = utf8
[mysqld]
character-set-server = utf8
```

Рис. 11: Содержимое файла utf8.cnf

Конфигурация кодировки. Перезапуск службы

```
[root@server ~]# systemctl restart mariadb
[root@server ~]# systemctl status mariadb
● mariadb.service - MariaDB 10.11 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Mon 2025-10-05 14:01:02 UTC; 7s ago
   Invocation: a3d67e9a07ab4dddbb915811b1f970956
   Docs: man:mariadb(8)
         https://mariadb.com/kb/en/library/systemd/
   Process: 12860 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)
   Process: 12882 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCESS)
   Process: 12919 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera_recovery ] && VAR= || VAR=galera_recovery (code=exited, status=0/SUCCESS)
   Process: 12932 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)
   Process: 12933 ExecStartPost=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)
```

Рис. 12: Перезапуск службы mariadb

Конфигурация кодировки. Статус после изменения

```
[root@server ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.11.18-MariaDB, for Linux (x86_64) using Editline wrapper

Connection id:          3
Current database:
Current user:           root@localhost
SSL:                    Not in use
Current pager:          stdout
Using outfile:          ''
Using delimiter:        ;
Server:                 MariaDB
Server version:         10.11.18-MariaDB MariaDB Server
Protocol version:       10
Connection:             Localhost via UNIX socket
Server characterset:    utf8mb3
Db characterset:        utf8mb3
Client characterset:    utf8mb3
Conn. characterset:     utf8mb3
UNIX socket:            /var/lib/mysql/mysql.sock
Uptime:                 28 sec

Threads: 1  Questions: 4  Slow queries: 0  Opens: 17  Open tables: 10  Queries per second avg: 0.142
-----
MariaDB [(none)]> |
```

Рис. 13: Вывод статуса MariaDB после изменения кодировки

```
[root@server ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 4
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE addressbook CHARACTER SET utf8 COLLATE utf8_general_ci;
Query OK, 1 row affected (0.003 sec)

MariaDB [(none)]> USE addressbook;
Database changed
```

Рис. 14: Создание базы данных addressbook

Создание базы данных. Таблица city

```
MariaDB [addressbook]> SHOW TABLES;
Empty set (0.006 sec)

MariaDB [addressbook]> CREATE TABLE city(name VARCHAR(40), city VARCHAR(40));
Query OK, 0 rows affected (0.120 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Иванов','Москва');
Query OK, 1 row affected (0.018 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Петров','Сочи');
Query OK, 1 row affected (0.018 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Сидоров','Дубна');
Query OK, 1 row affected (0.017 sec)

MariaDB [addressbook]> SELECT * FROM city;
+-----+-----+
| name | city |
+-----+-----+
| Иванов | Москва |
| Петров | Сочи |
| Сидоров | Дубна |
+-----+-----+
3 rows in set (0.005 sec)

MariaDB [addressbook]> |
```

Рис. 15: Создание таблицы city и заполнение её данными

Создание базы данных. Пользователь и права

```
MariaDB [addressbook]> CREATE USER mavinogradova@'%' IDENTIFIED BY 'password';
Query OK, 0 rows affected (0.060 sec)

MariaDB [addressbook]> GRANT SELECT, INSERT, UPDATE, DELETE ON addressbook.* TO mavinogradova@'%' ;
Query OK, 0 rows affected (0.019 sec)

MariaDB [addressbook]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.009 sec)

MariaDB [addressbook]> DESCRIBE city;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| name  | varchar(40)   | YES  |     | NULL    |       |
| city  | varchar(40)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.005 sec)

MariaDB [addressbook]> quit
Bye
[root@server ~]#
```

Рис. 16: Создание пользователя и выдача прав

Создание базы данных. Проверка через mysqlshow

```
[root@server ~]# mysqlshow -u root -p
Enter password:
+-----+
|      Databases      |
+-----+
| addressbook          |
| information_schema    |
| mysql                 |
| performance_schema    |
| sys                   |
+-----+

[root@server ~]# mysqlshow -u root -p addressbook
Enter password:
Database: addressbook
+-----+
| Tables |
+-----+
| city   |
+-----+

[root@server ~]# mysqlshow -u mavinogradova -p addressbook
Enter password:
Database: addressbook
+-----+
| Tables |
+-----+
| city   |
+-----+

[root@server ~]# |
```

Рис. 17: Просмотр списка баз данных и таблиц

Резервные копии. Обычная копия

```
[root@server ~]# mkdir -p /var/backup
[root@server ~]# ls -ld /var/backup
drwxr-xr-x. 2 root root 6 Oct  5 14:18 /var/backup
[root@server ~]# mysqldump -u root -p addressbook > /var/backup/addressbook.sql
Enter password:
[root@server ~]# ls -l /var/backup/addressbook.sql
-rw-r--r--. 1 root root 2007 Oct  5 14:18 /var/backup/addressbook.sql
[root@server ~]# head -20 /var/backup/addressbook.sql
/*M!999999\~ enable the sandbox mode */
-- MariaDB dump 10.19 Distrib 10.11.18-MariaDB, for Linux (x86_64)
--
-- Host: localhost    Database: addressbook
--
-- Server version      10.11.18-MariaDB
--
/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!40101 SET NAMES utf8 */;
/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
/*!40103 SET TIME_ZONE='+00:00' */;
/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;
--
-- Table structure for table `city`
[root@server ~]#
```

Рис. 18: Создание каталога /var/backup и обычной резервной копии

```
[root@server ~]# mysqldump -u root -p addressbook | gzip > /var/backup/addressbook.sql.gz
Enter password:
[root@server ~]# ls -l /var/backup/addressbook.sql.gz
-rw-r--r--. 1 root root 797 Oct 5 14:28 /var/backup/addressbook.sql.gz
[root@server ~]# mysqldump -u root -p addressbook | gzip > $(date + /var/backup/addressbook.%Y%m%d.%H%M%S.sql.gz)
date: extra operand '/var/backup/addressbook.%Y%m%d.%H%M%S.sql.gz'
Try 'date --help' for more information.
-bash: $(date + /var/backup/addressbook.%Y%m%d.%H%M%S.sql.gz): ambiguous redirect
Enter password:
mysqldump: Got errno 32 on write
[root@server ~]# mysqldump -u root -p addressbook | gzip > $(date + /var/backup/addressbook.%Y%m%d.%H%M%S.sql.gz)
Enter password:
[root@server ~]# ls -l /var/backup/
total 12
-rw-r--r--. 1 root root 797 Oct 5 14:22 addressbook.20261005.142156.sql.gz
-rw-r--r--. 1 root root 2887 Oct 5 14:18 addressbook.sql
-rw-r--r--. 1 root root 797 Oct 5 14:28 addressbook.sql.gz
[root@server ~]#
```

Рис. 19: Создание сжатых резервных копий

```
[root@server ~]# mysql -u root -p123456 -e "DROP TABLE addressbook.city;"
[root@server ~]# mysql -u root -p123456 -e "SHOW TABLES FROM addressbook;"
[root@server ~]# mysql -u root -p123456 -e "SHOW TABLES FROM addressbook;"
[root@server ~]# mysql -u root -p addressbook < /var/backup/addressbook.sql
Enter password:
[root@server ~]# mysql -u root -p123456 -e "SELECT * FROM addressbook.city;"
+-----+-----+
| name  | city  |
+-----+-----+
| Иванов | Москва |
| Петров | Сочи  |
| Сидоров | Дубна  |
+-----+-----+
[root@server ~]#
```

Рис. 20: Восстановление базы данных из несжатой резервной копии

```
[root@server ~]# mysql -u root -p123456 -e "DROP TABLE addressbook.city;"
[root@server ~]# zcat /var/backup/addressbook.sql.gz | mysql -u root -p addressbook
Enter password:
[root@server ~]# mysql -u root -p123456 -e "SELECT * FROM addressbook.city;"
+-----+-----+
| name  | city  |
+-----+-----+
| Иванов| Москва|
| Петров| Сочи  |
| Сидоров| Дубна |
+-----+-----+
[root@server ~]#
```

Рис. 21: Восстановление базы данных из сжатой резервной копии

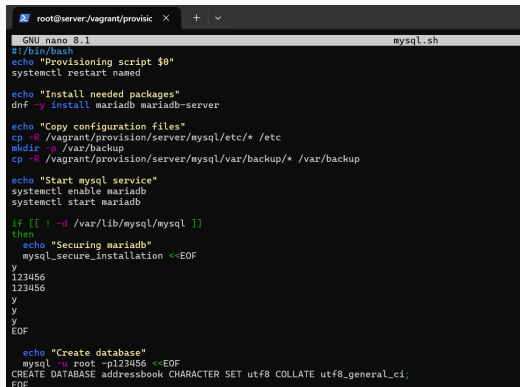
```
[root@server ~]# cd /vagrant/provision/server
[root@server server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d
[root@server server]# mkdir -p /vagrant/provision/server/mysql/var/backup
[root@server server]# cp -R /etc/my.cnf.d/utf8.cnf /vagrant/provision/server/mysql/etc/my.cnf.d/
[root@server server]# cp -R /var/backup/* /vagrant/provision/server/mysql/var/backup/
[root@server server]# find /vagrant/provision/server/mysql -type f
/vagrant/provision/server/mysql/etc/my.cnf.d/utf8.cnf
/vagrant/provision/server/mysql/var/backup/addressbook.20261005.142156.sql.gz
/vagrant/provision/server/mysql/var/backup/addressbook.sql
/vagrant/provision/server/mysql/var/backup/addressbook.sql.gz
[root@server server]#
```

Рис. 22: Копирование конфигурационного файла и резервных копий в каталог провизининга

```
[root@server server]# cd /vagrant/provision/server  
[root@server server]# touch mysql.sh  
[root@server server]# chmod +x mysql.sh  
[root@server server]# nano mysql.sh  
[root@server server]# |
```

Рис. 23: Создание исполняемого файла mysql.sh

Внесение изменений в окружение. Содержимое mysql.sh

A screenshot of a terminal window with a dark background. The window title is 'root@server:/vagrant/provisic'. The terminal shows the GNU nano 8.1 editor editing a file named 'mysql.sh'. The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
systemctl restart named

echo "Install needed packages"
dnf -y install mariadb mariadb-server

echo "Copy configuration files"
cp -R /vagrant/provision/server/mysql/etc/* /etc
mkdir -p /var/backup
cp -R /vagrant/provision/server/mysql/var/backup/* /var/backup

echo "Start mysql service"
systemctl enable mariadb
systemctl start mariadb

if [[ ! -d /var/lib/mysql/mysql ]]
then
    echo "Securing mariadb"
    mysql_secure_installation <<EOF
y
123456
123456
y
y
y
EOF

    echo "Create database"
    mysql -u root -p123456 <<EOF
CREATE DATABASE addressbook CHARACTER SET utf8 COLLATE utf8_general_ci;
EOF
```

Рис. 24: Содержимое скрипта провижининга mysql.sh

```
        preserve_order: true,  
        path: "provision/server/http.sh"  
  
    server.vm.provision "server mysql",  
        type: "shell",  
        preserve_order: true,  
        path: "provision/server/mysql.sh"  
end  
|
```

Рис. 25: Фрагмент Vagrantfile с добавленным блоком провижининга

Выводы

В ходе лабораторной работы были приобретены практические навыки по установке и конфигурированию СУБД MariaDB. Установлены пакеты `mariadb` и `mariadb-server`, служба включена в автозапуск и запущена. Настроена кодировка символов по умолчанию `utf8` через файл `utf8.cnf`. Создана тестовая база данных `addressbook` с таблицей `city` и тремя записями, создан пользователь с правами `SELECT, INSERT, UPDATE, DELETE` на эту базу. Освоены резервное копирование (несжатое, сжатое, с датой в имени) и восстановление базы данных. Написан скрипт провижининга `mysql.sh`, автоматизирующий установку и настройку MariaDB, и внесены соответствующие изменения в `Vagrantfile`.